

INDIAN STATISTICAL INSTITUTE, KOLKATA
ASSIGNMENT: FIRST SEMESTER 2024 -'25
M. MATH I

Subject: **Linear Algebra** Date: **September 12, 2024**
Duration: **2 hrs 30 mins** Maximum score: **30**

Attempt all the problems. Justify every step in order to get full credit of your answers. All arguments should be clearly mentioned on the answerscript. Points will be deducted for missing or incomplete arguments.

- (1) Let V be a finite dimensional vector space and $T \in L(V, V)$. Show that $T^2 = T$ (that is, T is idempotent) **if and only if** $\text{rank}(T) + \text{rank}(\text{id}_V - T) = \dim(V)$.

[8 marks]

- (2) For $n \in \mathbb{N}$, consider the continuous function $f_n : [0, 1] \rightarrow \mathbb{R}$ defined by $f_n(t) := \max\{0, (1 - nt)\}$. Prove or disprove (with justification) that $\{f_n : n \in \mathbb{N}\}$ is a linearly independent subset of $C[0, 1]$.

[6 marks]

- (3) (a) Let $A \in M_n = M_n(\mathbb{R})$ such that $AA^t = 1_{M_n}$ and $\det(A) < 0$. Find $\det(A + 1_{M_n})$.
(b) Let $X, Y \in M_4(\mathbb{R})$ such that $\det(X) = 7$, $\det(Y) = 2$ and $c_j(X) = c_j(Y)$ for all $j = 2, 3, 4$. Deduce $\det(X + Y)$ and $\det(Z)$ where $Z := (c_4(X), c_3(X), c_2(X), c_1(X) + c_1(Y)) \in M_4$.

[8 marks]

- (4) Let V be an n dimensional vector space, $\{b_j\}_{j=1}^n$ be a basis for V and $\{b_j^*\}_{j=1}^n$ be the dual basis for $V^\#$ with respect to $\{b_j\}_{j=1}^n$.

(a) If $\{c_j\}_{j=1}^n$ is another basis and $X \in M_n$ is the transfer matrix from $\{b_j\}_{j=1}^n$ to $\{c_j\}_{j=1}^n$ (that is, $b_j = \sum_{k=1}^n X_{k,j} c_k$ for all j), then find the transfer matrix from $\{b_j^*\}_{j=1}^n$ to $\{c_j^*\}_{j=1}^n$. Hint: Argue that X is invertible and work with its inverse.

(b) Show that $\sum_{k=1}^n (b_k \otimes b_k^*) \in V \otimes V^\#$ is independent of the choice of basis $\{b_j\}_{j=1}^n$.

[8 marks]

- (5) Let V be a finite dimensional vector space over $\mathbb{R}$ and $T \in L(V, V)$ such that $m_T(x) = (x^2 + 4x + 15)(x + 7)$. Explicitly find $p \in \mathbb{R}[x]$ such that $p(T)$ is an idempotent in $L(V, V)$ satisfying $\text{Range}(p(T)) = \ker(T^2 + 4T + 15\text{id}_V)$.

[8 marks]

Handwritten work for problem (5):

Let $m_T(x) = (x^2 + 4x + 15)(x + 7)$. We want to find $p(x)$ such that $p(T)$ is an idempotent and $\text{Range}(p(T)) = \ker(T^2 + 4T + 15\text{id}_V)$.

Let $p(x) = \frac{x^2 + 4x + 15}{x^2 + 4x + 15 + (x + 7)}$. Then $p(T)$ is an idempotent and $\text{Range}(p(T)) = \ker(T^2 + 4T + 15\text{id}_V)$.

Alternatively, let $p(x) = \frac{x^2 + 4x + 15}{x^2 + 4x + 15 + (x + 7)}$. Then $p(T)$ is an idempotent and $\text{Range}(p(T)) = \ker(T^2 + 4T + 15\text{id}_V)$.